Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building **C++/Python-adjacent systems**, robotics evaluation pipelines, telemetry-style logging, CI/CD validation, and reliable production software.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **Python-backed observability workflows** for 500+ users, translating requirements into APIs, validation logic, monitored jobs, technical documentation, release notes, and maintainable delivery through reviewed changes and support runbooks.
- Improved **backend reliability and system performance** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** with diagnostic handoffs and reviews.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **Python services and React tooling** handling 10,000+ daily requests, implementing file ingestion, automated checks, documented fixes, and sprint-ready production workflows with code review and monitoring.
- Debugged **production workflow issues** by reproducing OCR, persistence, and execution defects, documenting regression cases, reviewing fixes, and cutting instructor feedback turnaround by **60%** across validation cycles and releases.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **autonomous-driving evaluation pipelines** across 100,000+ trajectory records, producing metrics, logs, and reports for simulation review, model behavior debugging, planning validation, experiment handoffs, and sensor-style analysis.
- Integrated **HD-map geometry and trajectory context** with attention-based encoders, improving planning accuracy by **18%** while surfacing edge cases through closed-loop evaluation, diagnostics, simulation reviews, and validation reports.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **cross-platform API and ETL services** using Python, SQL, AWS Glue, and S3, improving response time by **30%** across integrations, releases, reviews, handoffs, operational checks, and telemetry-style workflows.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito across validation flows, increasing CI reliability by **68%** and reducing release regressions during deployments, checks, and reviews.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **real-time application modules** with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting 2M+ synchronized records across dashboards, workflows, integrations, reviews, releases, debugging sessions, validation, and runtime logs.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting documentation, validation notes, and shared-environment handoffs across staging workflows.

Skills

Languages: Python, C++, C familiarity, Java, JavaScript, TypeScript, SQL, Bash
Robotics: Linux/Ubuntu, telemetry, logging, observability, sensor-data concepts, middleware concepts, HAL awareness
Systems: data structures, algorithms, OOP, multithreading concepts, performance profiling, CPU/memory/latency awareness
Backend: REST APIs, Flask, FastAPI, data ingestion, backend services, integration testing, production debugging
Tools: Git, Docker, Jenkins, CI/CD, PyTest, Mockito, PyTorch, PostgreSQL, MySQL, MongoDB

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ scheduling** with metric calculation and visualization, strengthening C++ fundamentals, debugging discipline, runtime reasoning, correctness testing, scheduler comparisons, and performance analysis.

CaptionCraft Live | *Transformers, PyTorch, Flask, Computer Vision*

- Benchmarked **computer-vision inference APIs** across BLIP, GIT, and ViT-GPT-2 models on 10K images and 100K captions, improving caption accuracy by **15%** with diagnostics, validation logs, and reproducible tests.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Developed a **containerized Python REST API** with AWS deployment, CI/CD releases, validation checks, and reproducible execution, achieving **87%** classification accuracy with diagnostic reporting, integration notes, and automated tests.